



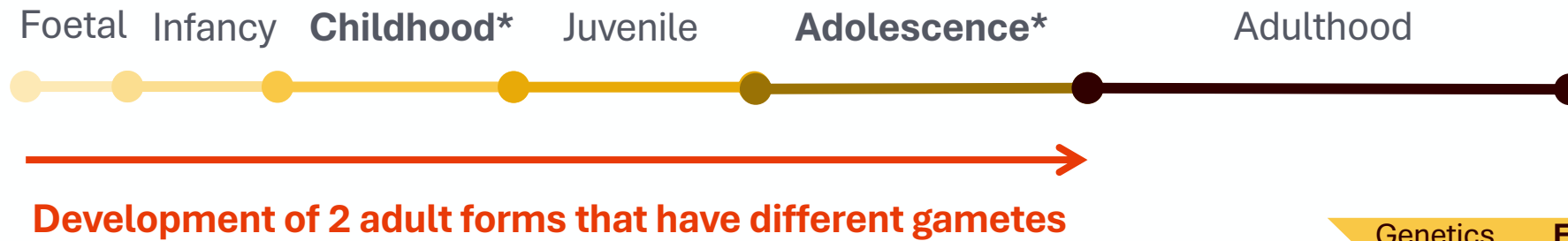
The Kiss (Klimt, 1908)

Reproduction II

Dr. Laura van Holstein

lavanho@emory.edu

ANT215

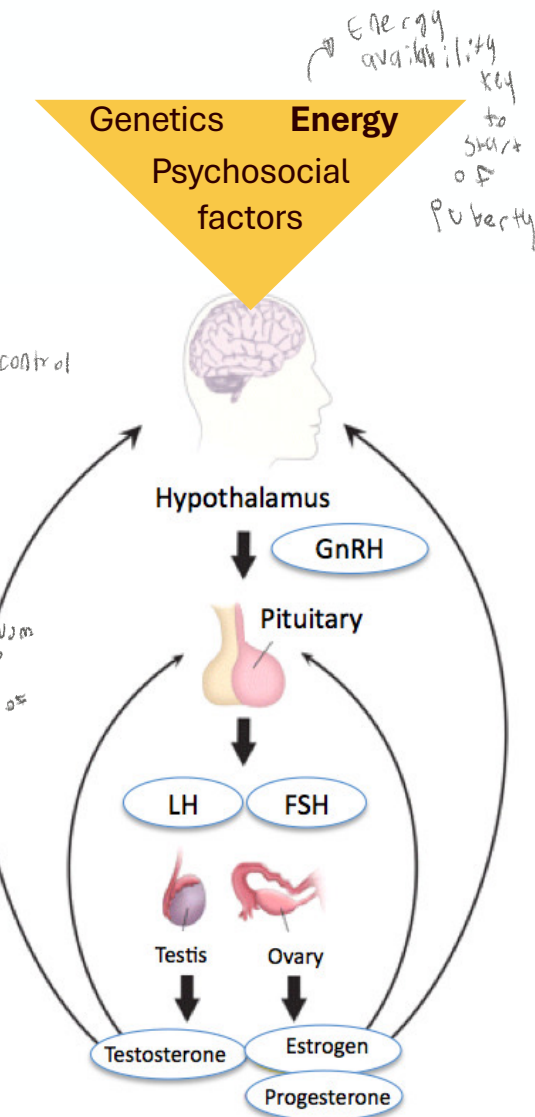


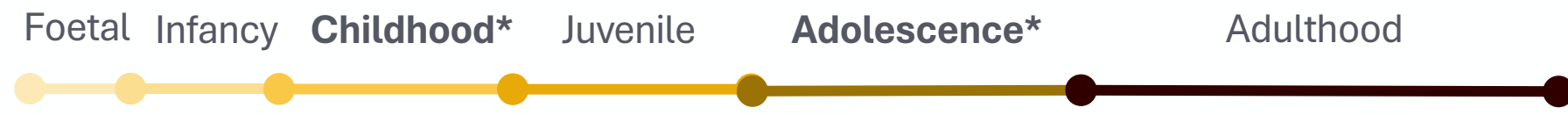
What is sex?

1. Chromosomal, gonadal, endocrine, genital, and phenotypic sex
2. There are strong correlations between these, but $R^2 \neq 1$! → Correlation exists due to strong endocrine control
3. There is no absolute binary at any of these levels apart from
 - the existence of the sex chromosomes (ish) X and Y, but can be different combo
 - Gametes (no spergs or speggs) – but some people could *produce* both

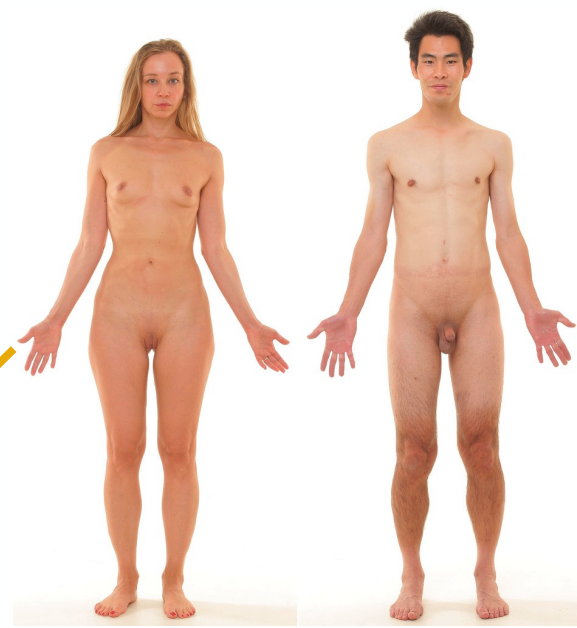
How does sex develop?

1. Chromosomal sex is set at fertilisation. The SRY gene on the Y-chromosome plays a key role in driving male differentiation
2. Followed by 6-week *indifferent period*. Some structures are bipotential (gonads and external genitalia); others aren't and *degrade* depending on SRY gene activation (Müllerian & Wolffian ducts)
3. 6+ weeks: sexual differentiation, driven largely by SRY – testes – T in males.
4. Minipuberty in infancy
5. Adolescence: divergent somatic & sexual growth patterns

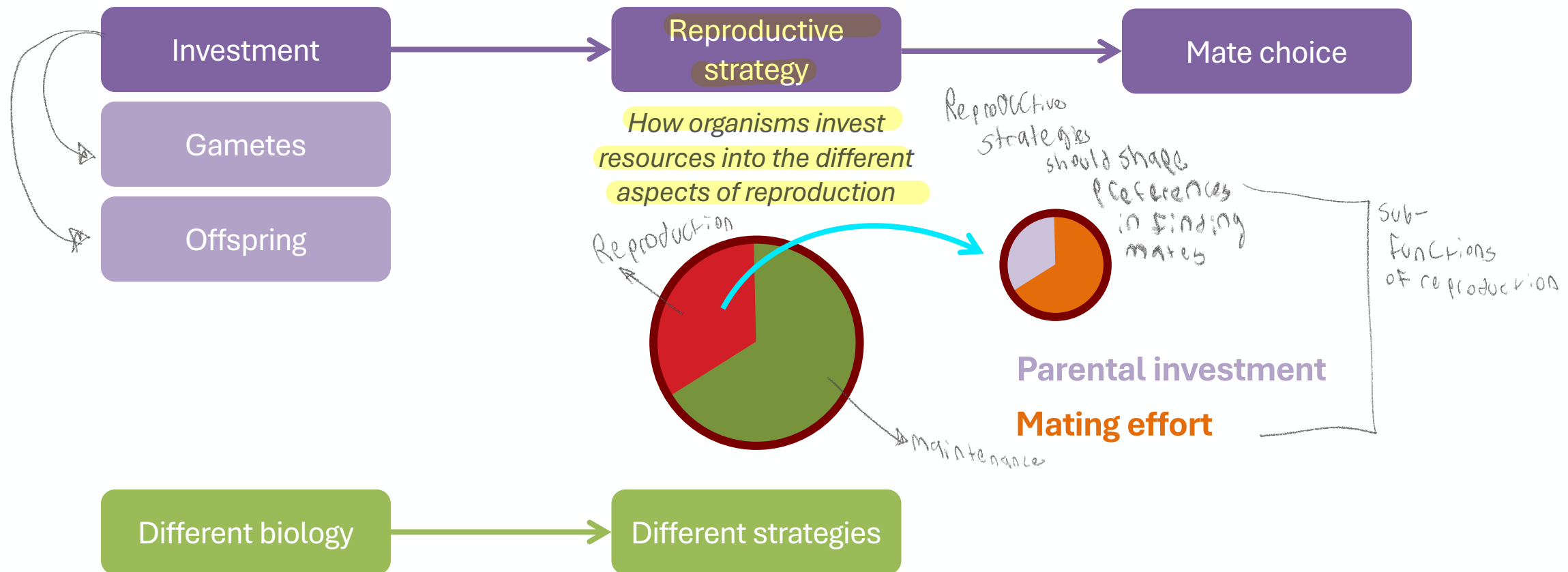


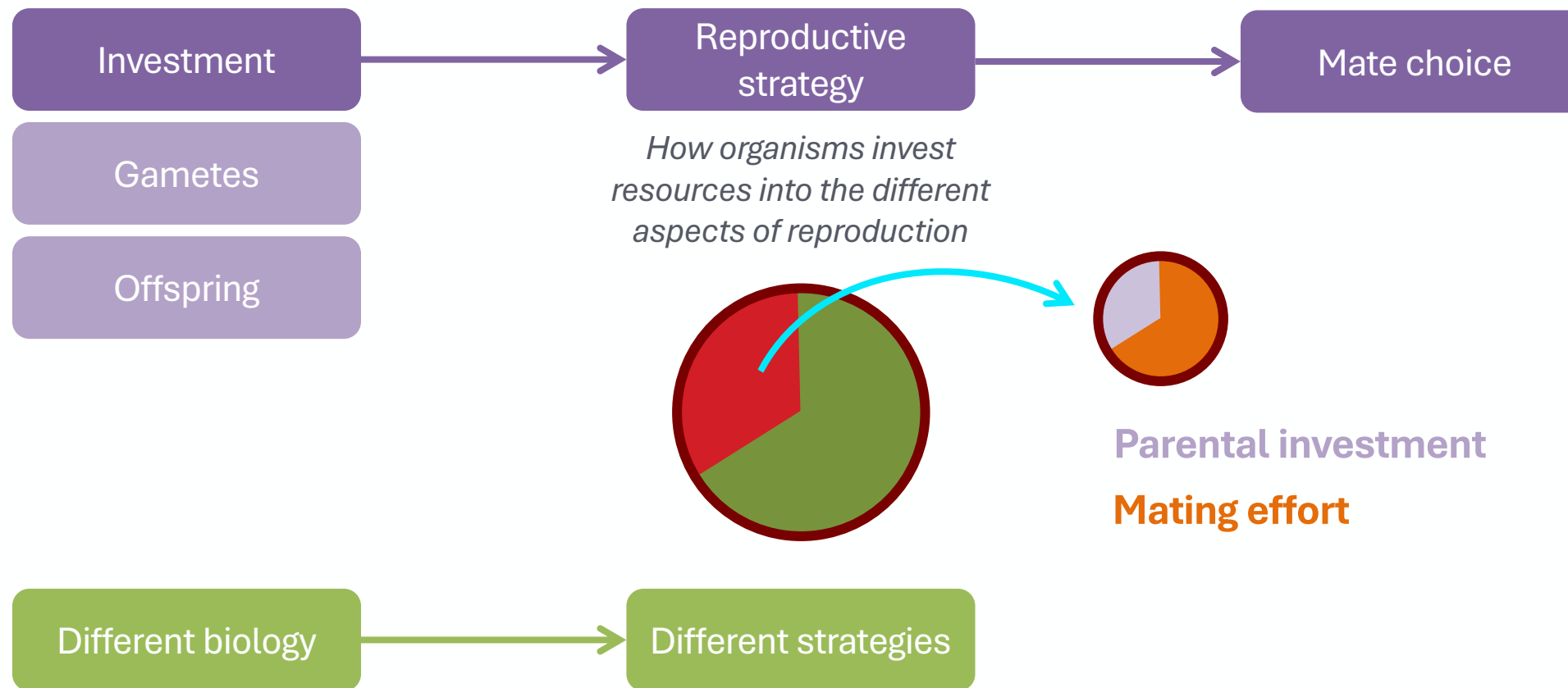


Development of 2 adult forms that have different gametes



Goal as a biological organism



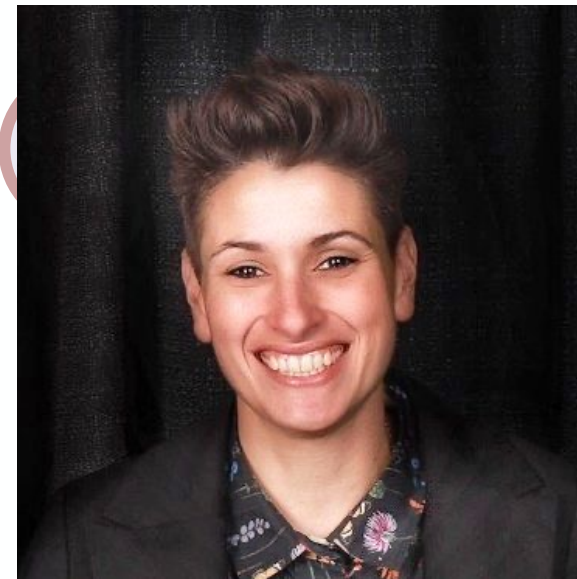
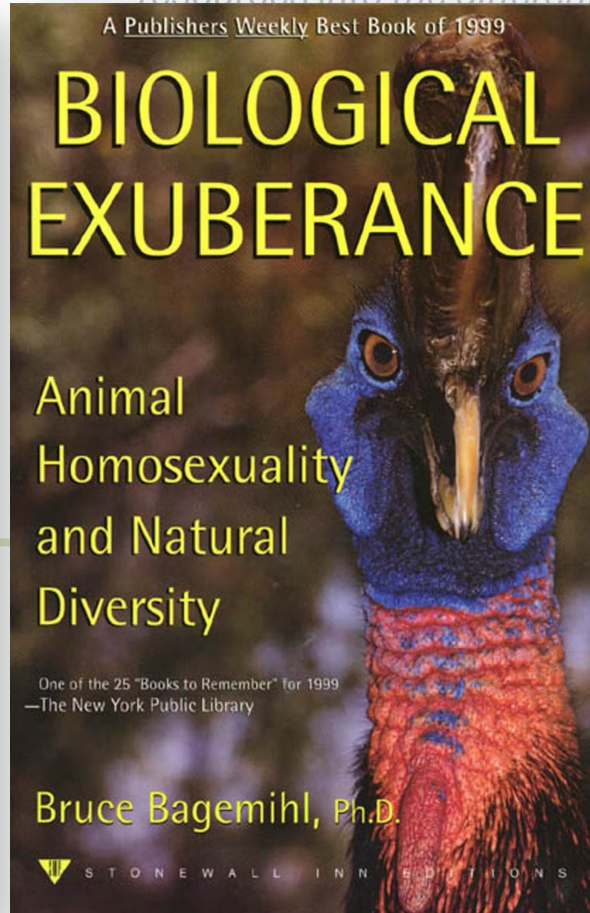
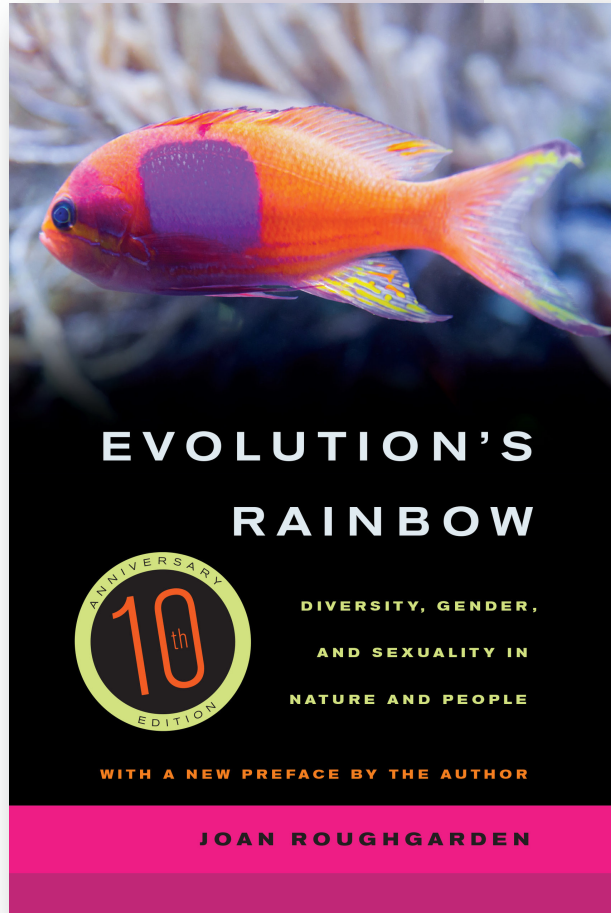


Investment

Reproductive
strategy

Mate choice

Gametes

*How organisms invest
resources into the different*

Prof. Benítez: *Nature, Nurture, Sexuality* (may be offered in 2026-27; keep an eye out on Course Atlas)

Homosexuality
is fairly
common
in biology

Investment

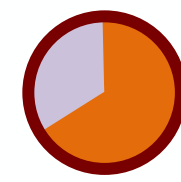
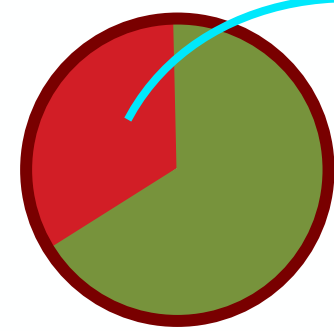
Gametes

Offspring

Reproductive strategy

How organisms invest resources into the different aspects of reproduction

Mate choice



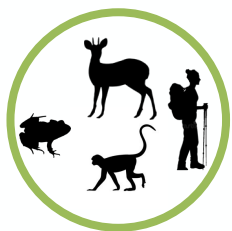
Parental investment

Mating effort

Different biology

Different strategies

many things we are talking about perhaps
All sexually reproducing organisms



Mammals



Humans

Do humans behave as expected for a mammal in their ecosystems?



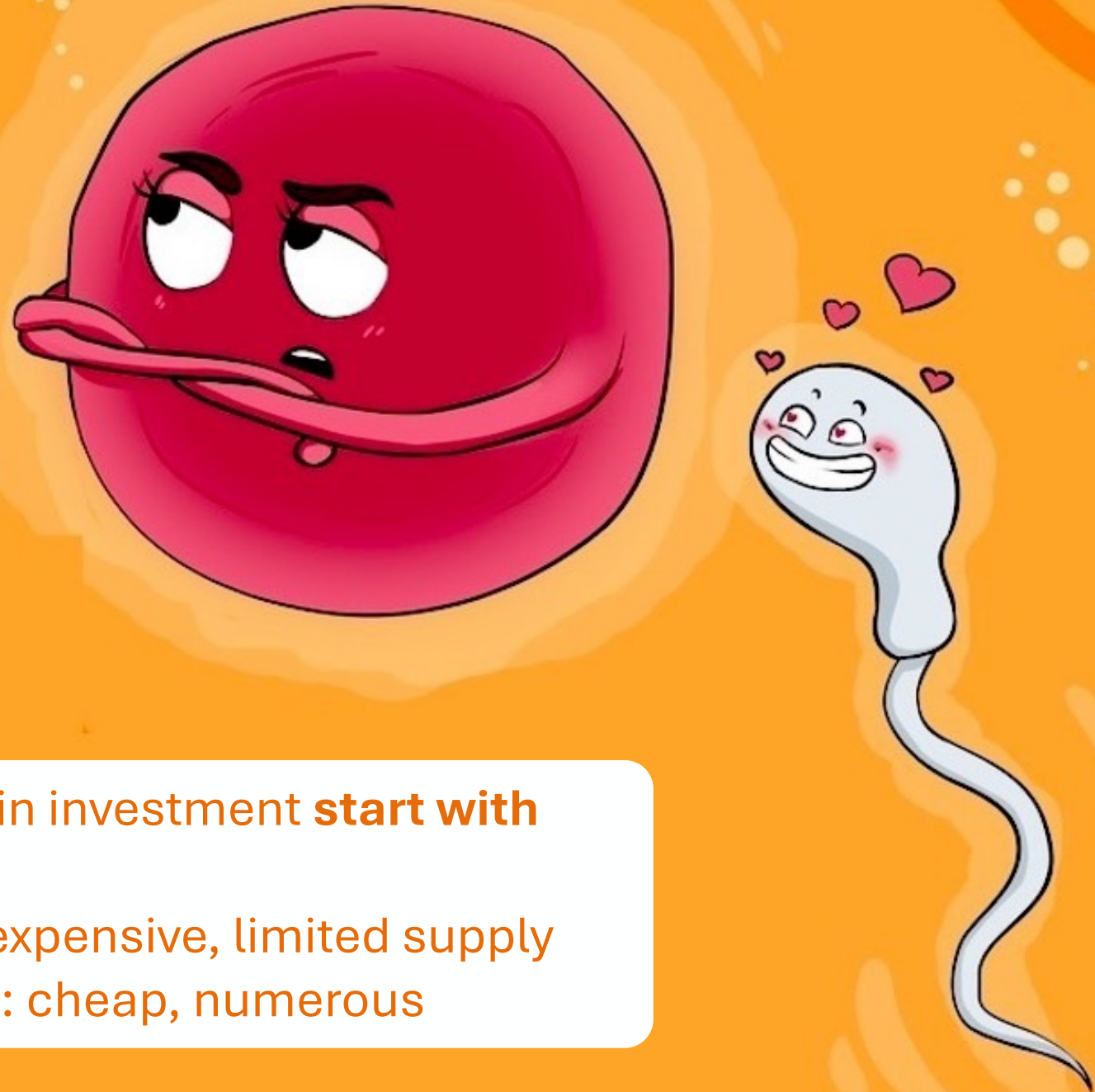


Anisogamy

Differences in the size and form of gametes

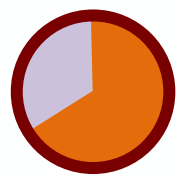
Oogamy

Large non-motile eggs; small, motile sperm



Differences in investment **start with gametes**

- Eggs: expensive, limited supply
- Sperm: cheap, numerous



Parental investment vs Mating effort

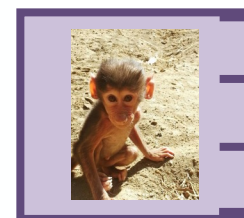
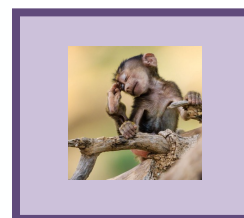
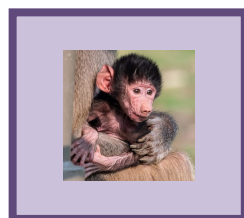
- Parental investment (Trivers 1972):

What is the overall effort
that you're
putting
into
your
off-
spring?



I first define **parental investment** as any investment by the parent in an individual offspring that increases the offspring's chance of surviving (and hence reproductive success) at the cost of the parent's ability to invest in other offspring. So defined, parental investment includes the meta-

↳ Loads of



putting a ton of
effort into babies

Large egg

Internal gestation
↳ mammal

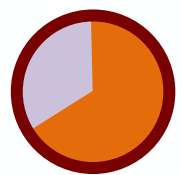
Lactation
↳ mammal

Provisioning



Sperm

↳ In most
primates, paternal
investment is just
sperm



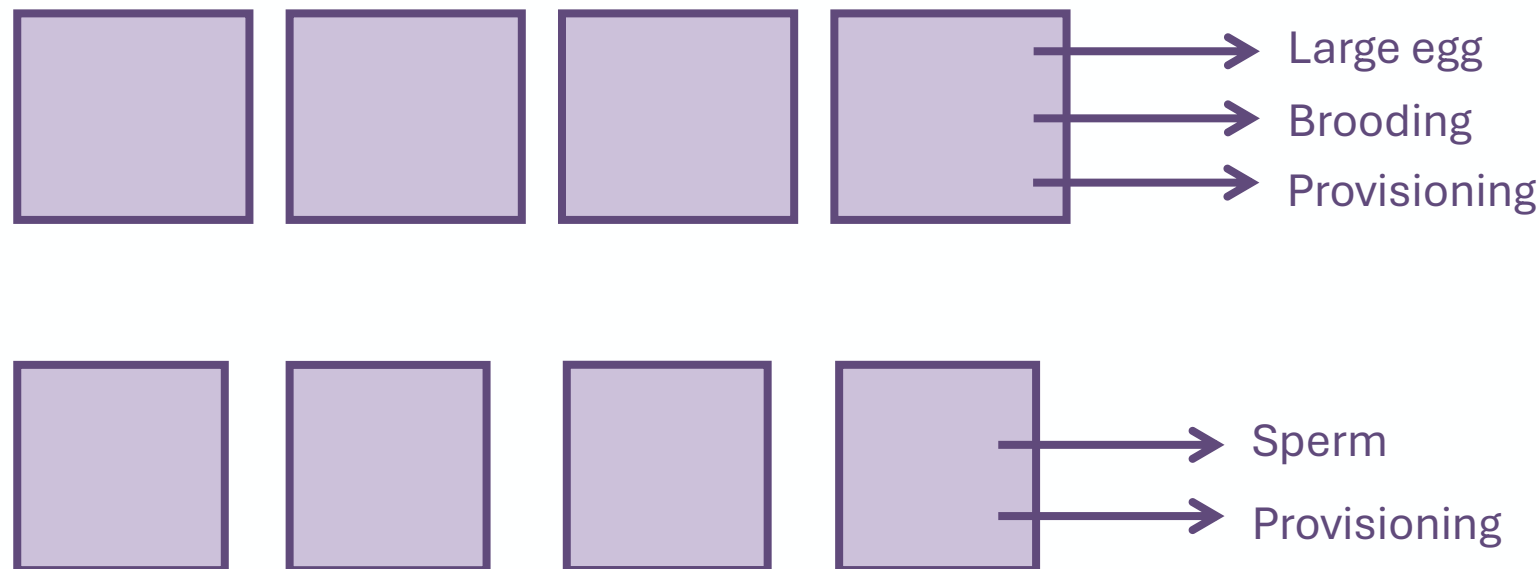
Parental investment vs Mating effort

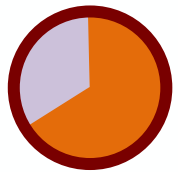
- **Parental investment** (Trivers 1972):

lection. I first define parental investment as *any investment by the parent in an individual offspring that increases the offspring's chance of surviving (and hence reproductive success) at the cost of the parent's ability to invest in other offspring*. So defined, parental investment includes the meta-

vs in birds ...

Roughly equal investment



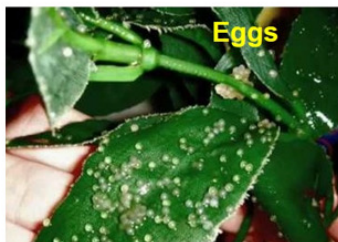


Parental investment vs Mating effort

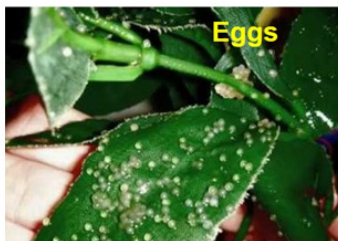
- Parental investment (Trivers 1972):

In fish, roughly equal investment

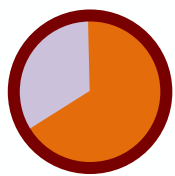
I first define parental investment as *any investment by the parent in an individual offspring that increases the offspring's chance of surviving (and hence reproductive success) at the cost of the parent's ability to invest in other offspring*. So defined, parental investment includes the meta-



Large egg



Sperm



Parental investment vs Mating effort

- **Parental investment** (Trivers 1972):

lection. I first define parental investment as *any investment by the parent in an individual offspring that increases the offspring's chance of surviving (and hence reproductive success) at the cost of the parent's ability to invest in other offspring*. So defined, parental investment includes the meta-

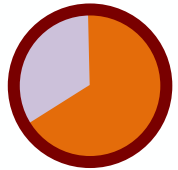
Maynard Smith (1977): ESS model of parental investment

Costs of trade-offs for both sexes lead to 4 strategies:

1. **Both desert**
2. **Female care** (if male can increase reproductive success by mating again, and one parent is much better than none)
3. **Male care** (if female can lay more eggs if she does not invest in caring, and one parent is much better than none)
4. **Biparental care** (if two parents can raise > twice as many young as one or if chance that deserting parent remates is small)

Not needed for exam

↳ mammals don't have all these options



Parental investment vs Mating effort

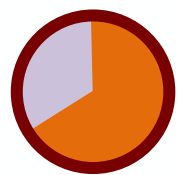
- **Parental investment** (Trivers 1972):

lection. I first define parental investment as *any investment by the parent in an individual offspring that increases the offspring's chance of surviving (and hence reproductive success) at the cost of the parent's ability to invest in other offspring*. So defined, parental investment includes the meta-

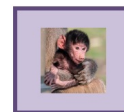
Maynard Smith (1977): ESS model of parental investment
Costs of trade-offs for both sexes lead to 4 strategies:

1. **Both desert**
2. **Female care** (if male can increase reproductive success by mating again, and one parent is much better than none) ↳ More limited in mammals
3. **Male care** (if female can lay more eggs if she does not invest in caring, and one parent is much better than none)
4. **Biparental care** (if two parents can raise > twice as many young as one or if chance that deserting parent remates is small)

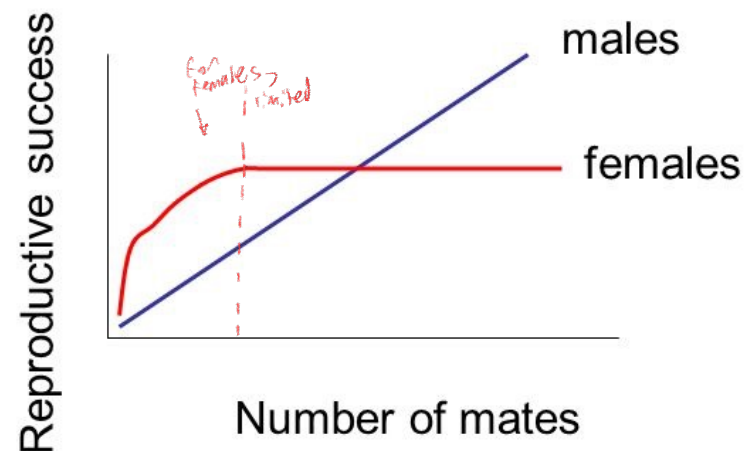




Parental investment

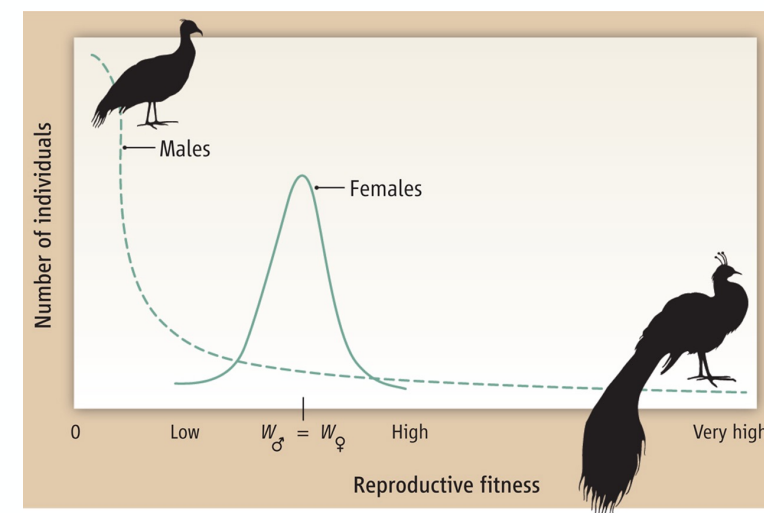


Bateman's principle (1948):



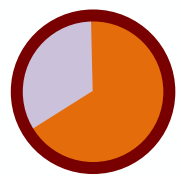
Clutton-Brock & Vincent (1991):

“the *potential* reproductive rate of males usually exceeds that of females”



Bateman, Angus J. "Intra-sexual selection in *Drosophila*." *Heredity* 2.3 (1948): 349-368.

Clutton-Brock, T. H., and Amanda CJ Vincent. "Sexual selection and the potential reproductive rates of males and females." *Nature* 351.6321 (1991): 58-60

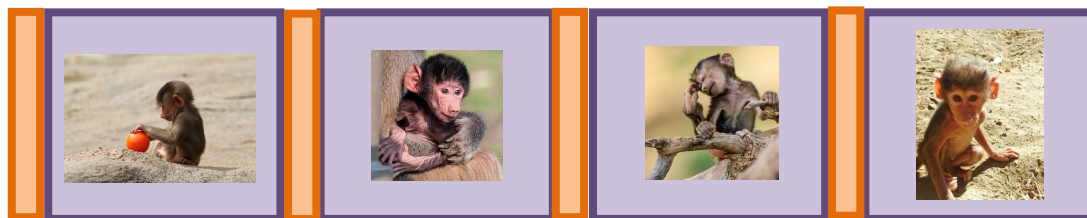


Parental investment vs Mating effort

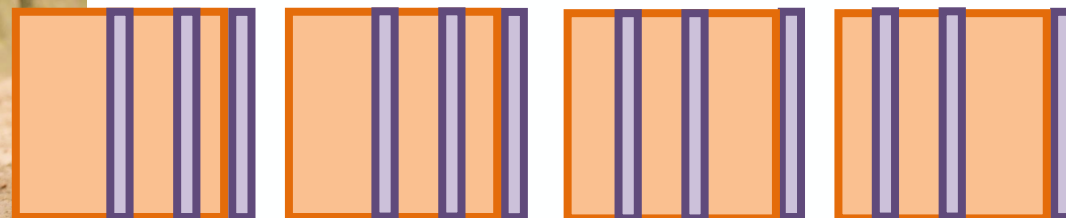
- **Parental investment** (Trivers 1972): all investment in individual offspring
- **Mating effort**: all investment in finding mates

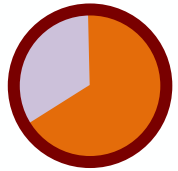


↳ Sex difference in energy investment for mates



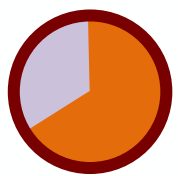
Doesn't have to be monogamous





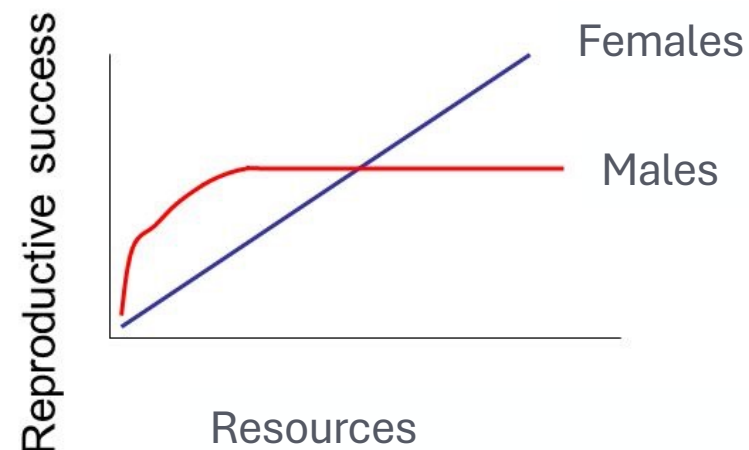
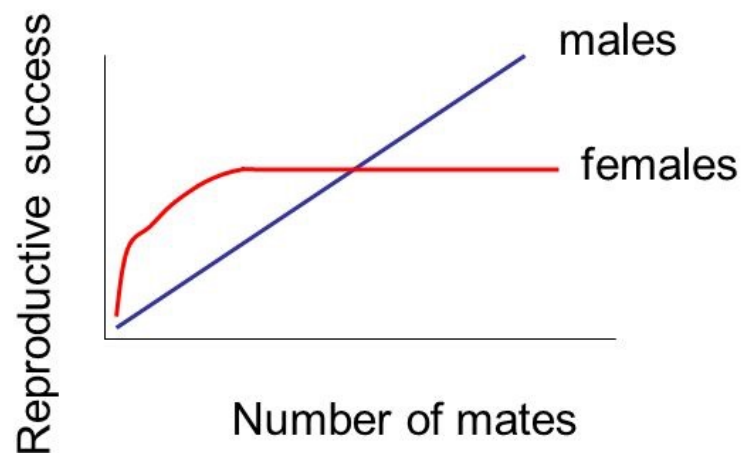
Parental investment vs Mating effort

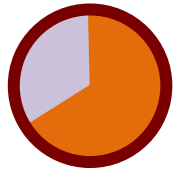
	Males	Females
Gametes	Low	High
Parental investment	Low	Often high



Parental investment vs Mating effort

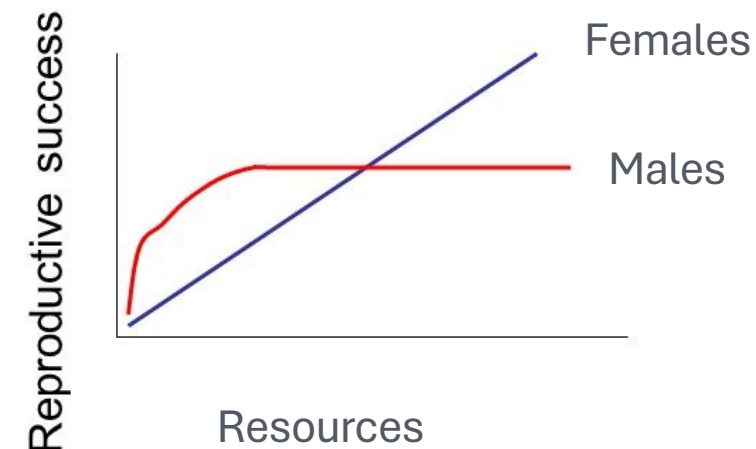
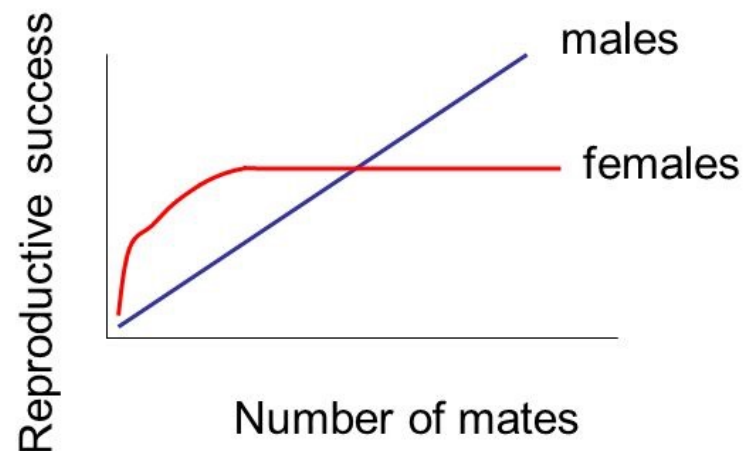
	Males	Females
Gametes	Low	High
Parental investment	Low	Often high
Limiting factors	Females	Resources

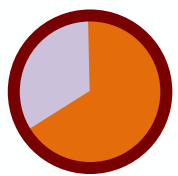




Parental investment vs Mating effort

	Males	Females
Gametes	Low	High
Parental investment	Very low	Very high
Limiting factors	Females	Resources





Parental investment vs Mating effort

	Males	Females
Gametes	Low	High
Parental investment	Very low	Very high
Limiting factors	Females	Resources



Expectations:

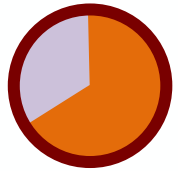
- Strategy focused on maximizing access to females
- Low parental investment; high mating effort
- High reproductive skew



Expectations:

- Strategy focused on maximizing access to resources
- High parental investment; low mating effort
- Low reproductive skew

Generally same # of offspring for same resources because



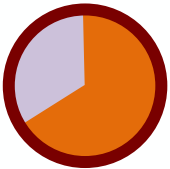
Parental investment vs Mating effort

Foetal Infancy **Childhood*** Juvenile **Adolescence***



	Males	Females
Gametes		
Parental investment		
Limiting factors		



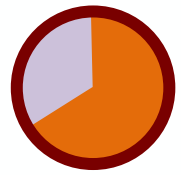


Parental investment vs Mating effort



	Males	Females
Gametes		
Parental investment		
Limiting factors		



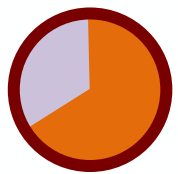


Parental investment vs Mating effort



	Males	Females
Gametes	Low	High
Parental investment	Comparatively high	Very high
Limiting factors	Females and resources	Resources





Parental investment vs Mating effort

Foetal Infancy **Childhood*** Juvenile **Adolescence***



	Males	Females
Gametes	Low	High
Parental investment	Comparatively high	Very high
Limiting factors	Females and resources	Resources



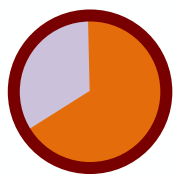
Expectations:

- Strategy focused on maximizing access to females *and* resources
- More equal balance between parental investment and mating effort
- High reproductive skew



Expectations:

- Strategy focused on maximizing access to resources
- High parental investment; low mating effort
- Low reproductive skew



Parental investment vs Mating effort

Foetal Infancy **Childhood*** Juvenile **Adolescence***



	Males	Females
Gametes	Low	High
Parental investment	Comparatively high	Very high
Limiting factors	Females and resources	Resources



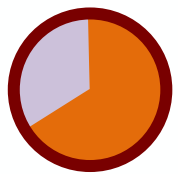
Expectations:

- Strategy focused on maximizing **access to females** and resources
- More equal balance between parental investment and **mating effort**
- High reproductive skew



Expectations:

- Strategy focused on maximizing access to resources
- High parental investment; **low mating effort**
- Low reproductive skew



Parental investment vs Mating effort

Where we fit in: Clark & Hatfield (1989)

Table 1: Study #1, 1978

Percentage of Compliance With Each Request

Sex of Requestor	Type of Request		
	Date	Apartment	Bed
Male	56%	6%	0%
Female	50%	69%	75%

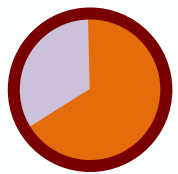
Women willing to go on date but not sex

More males want to have sex than go on a date

Focused on access to mating

More focused on long term





Parental investment vs Mating effort

Foetal Infancy **Childhood*** Juvenile **Adolescence***



	Males	Females
Gametes	Low	High
Parental investment	Comparatively high	Very high
Limiting factors	Females and resources	Resources



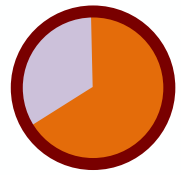
Expectations:

- Strategy focused on maximizing access to females *and* resources
- More equal balance between parental investment and mating effort
- **High reproductive skew**



Expectations:

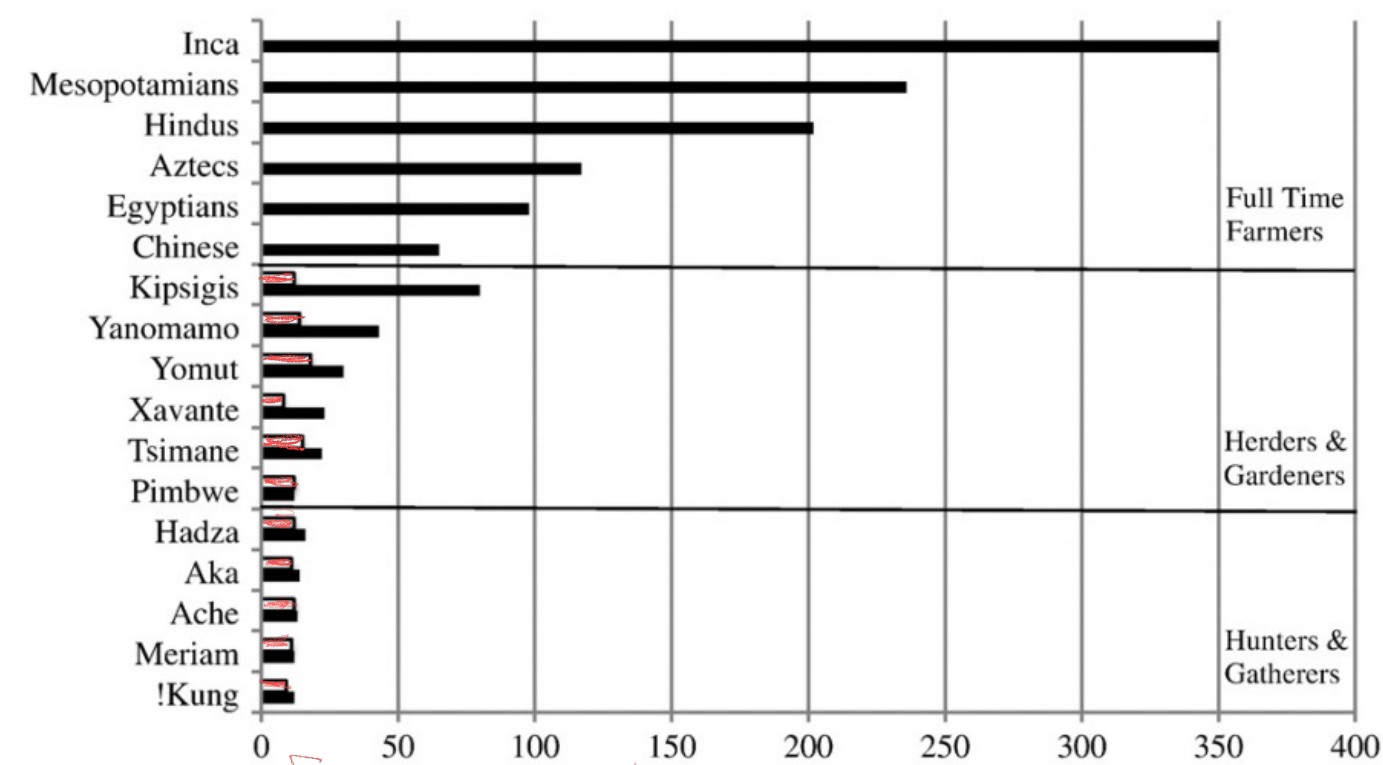
- Strategy focused on maximizing access to resources
- High parental investment; low mating effort
- **Low reproductive skew**



Parental investment vs Mating effort

Reproductive skew

Range of reproductive success:



Sedentism

Inequality

Resource monopolisability

Polygamy

Monogamy

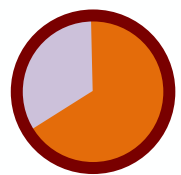
Females, generally abt 6 offspring H/G males have generally equal reproductive success to females

Intro

Reproductive strategies

Mate choice

Other

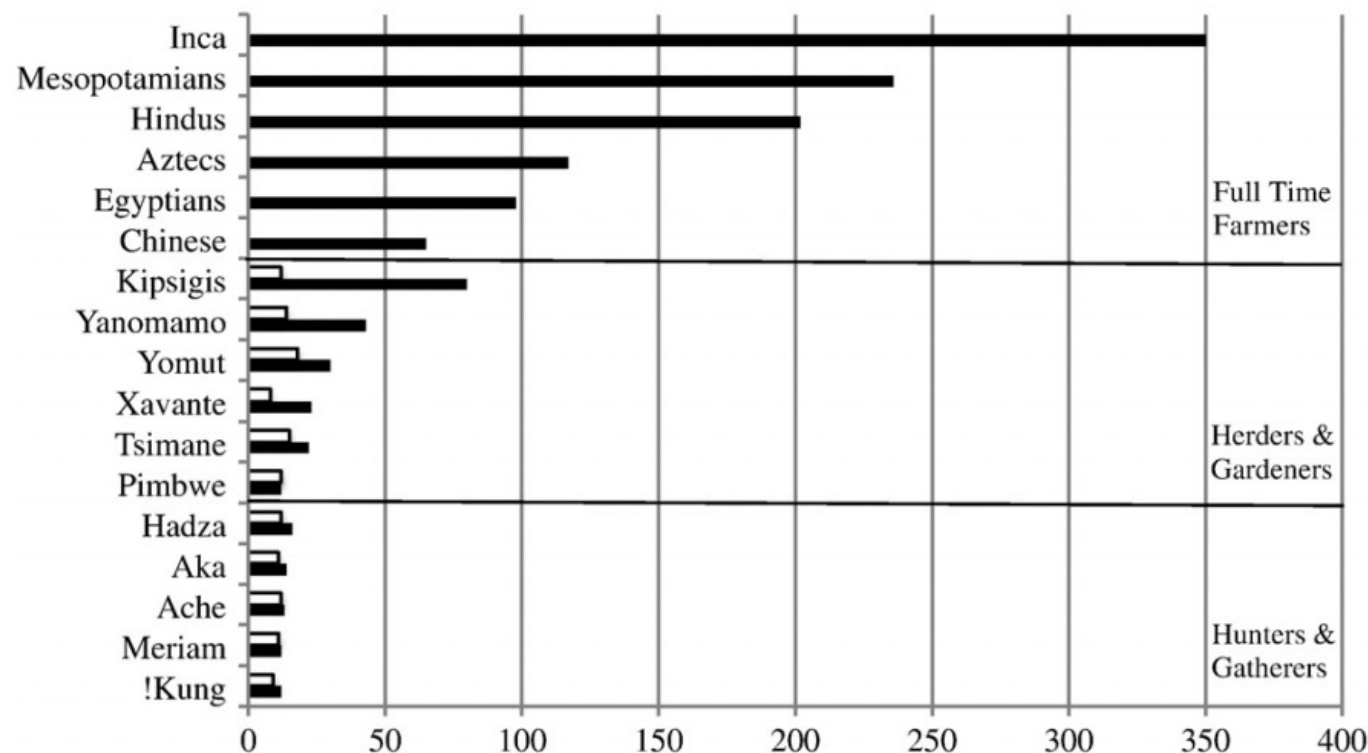


Parental investment vs Mating effort

Reproductive skew

The degree to which humans meet expectations follows social ecology

Range of reproductive success:



Ability to access resources & socially stratified

Sedentism

Inequality

Resource monopolisability

Polygamy

Monogamy

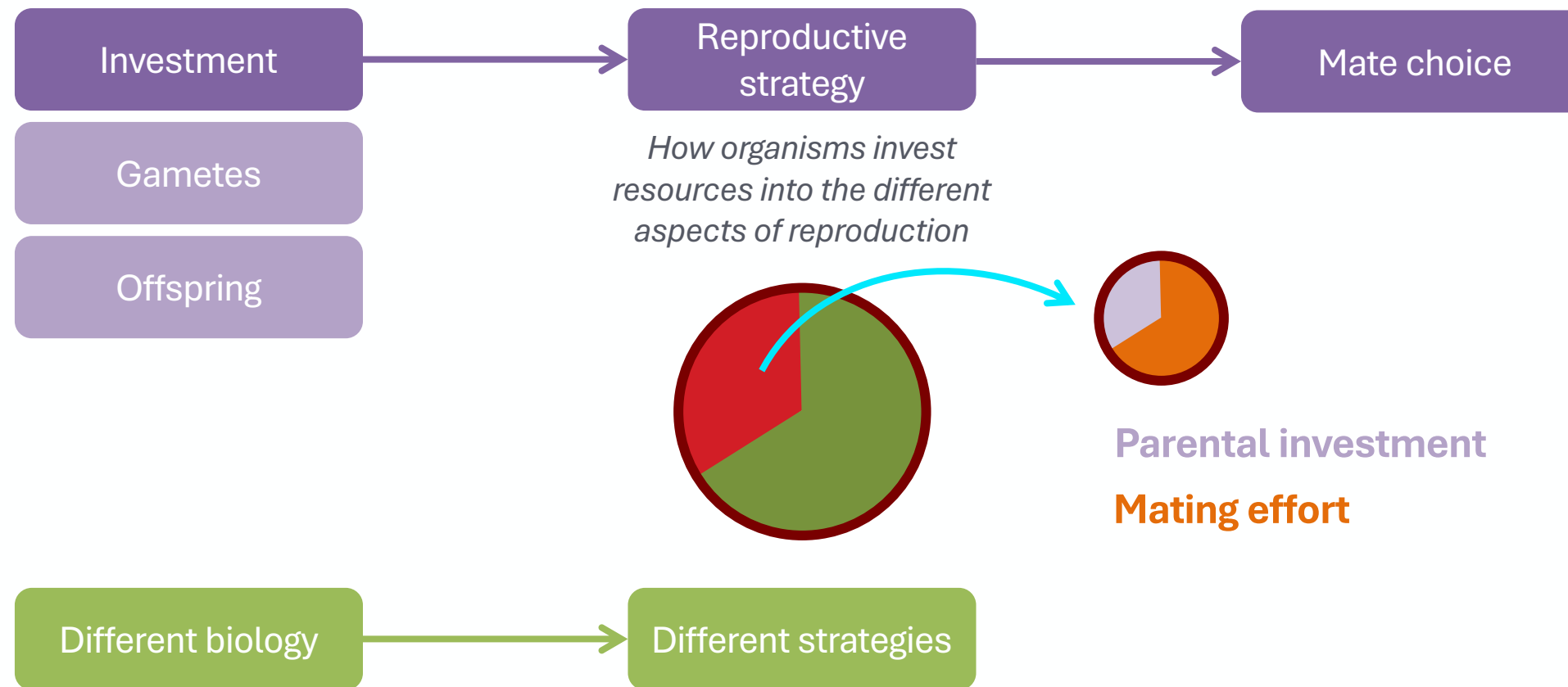
Key

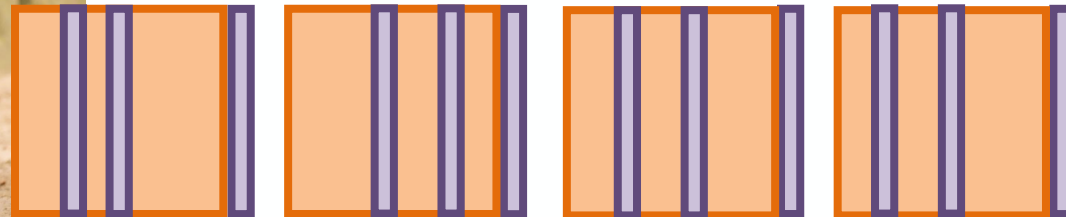
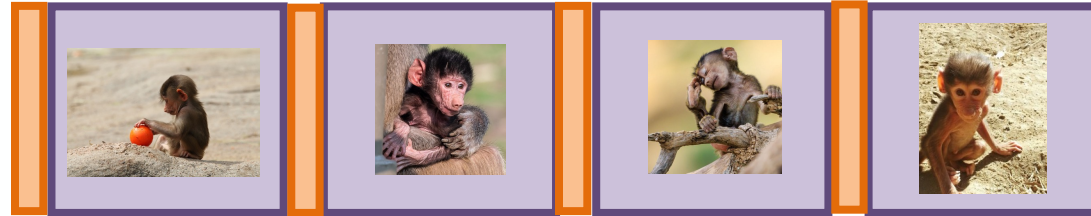
Intro

Reproductive strategies

Mate choice

Other





Ensure access to resources, and...

Blimey: I'd better make sure this is a success

Key

Females have fewer offspring
huge cost if unsuccessful

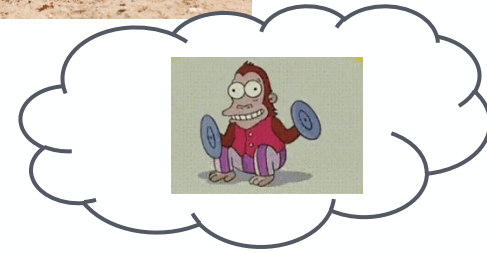
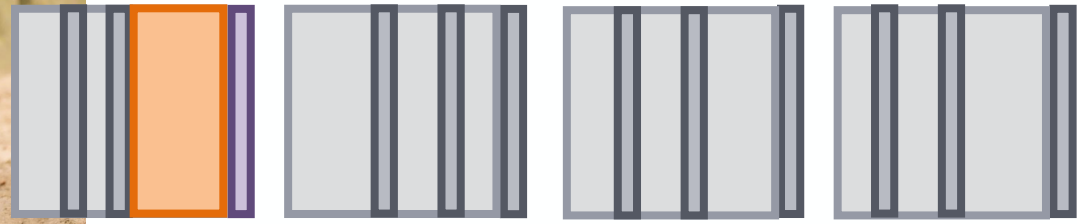
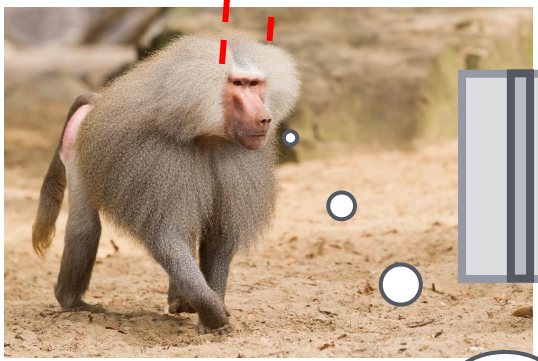
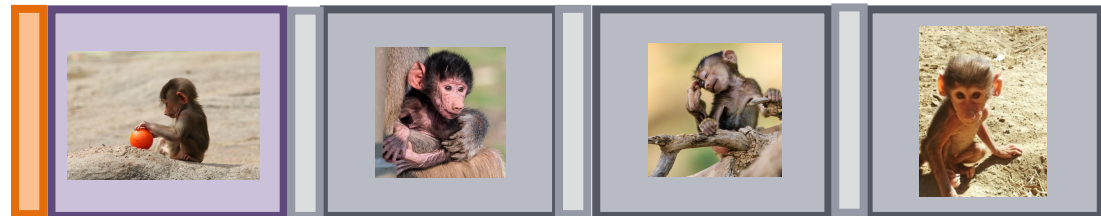
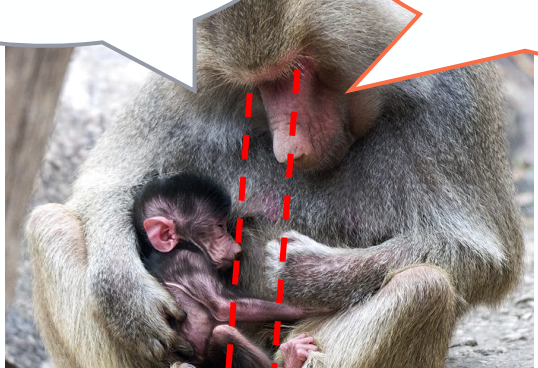
Success of a single reproductive event +

1/4 of lifetime fitness

Huge cost if unsuccessful

1/12 of lifetime fitness

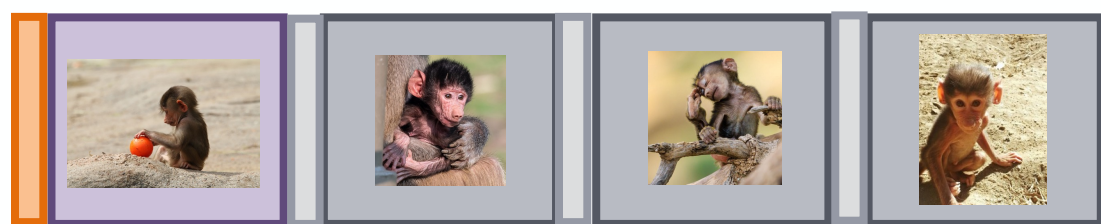
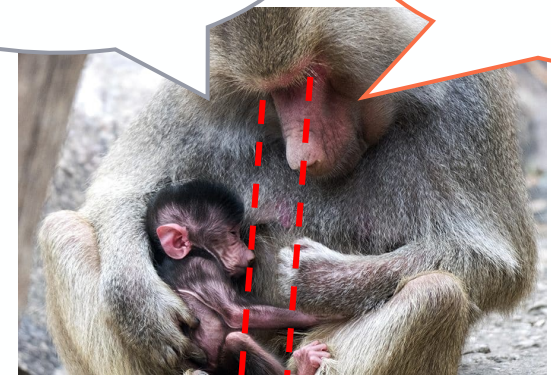
Low cost if unsuccessful



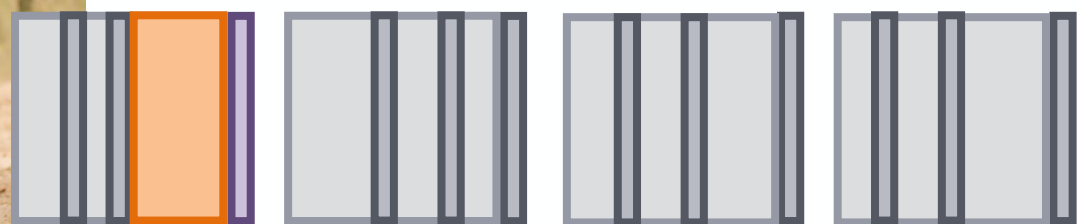
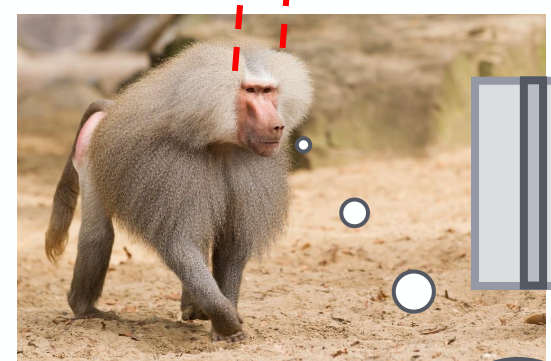
Ensure access
to resources,
and...

Blimey: I'd better
make sure this is a
success

Key



**1/4 of
lifetime
fitness**
Huge cost if
unsuccessful



**1/12 of
lifetime
fitness**
Low cost if
unsuccessful

*"Chicks for
free!"*

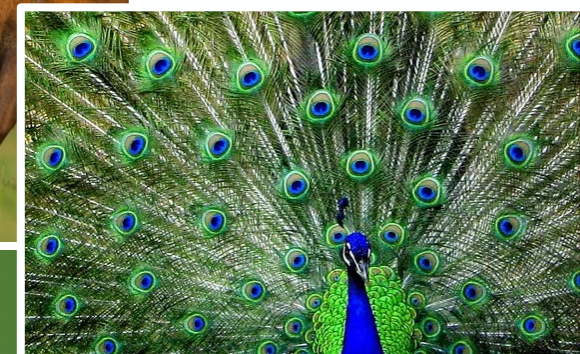
Because I want to reach
that high potential
reproductive success



Male reproductive strategy & mate preference

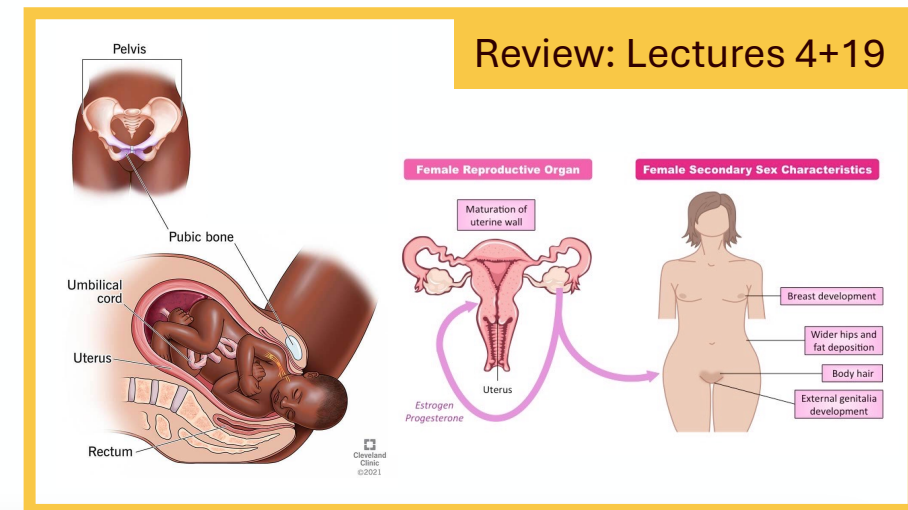
- Low investment & high **potential** reproductive rate: high mating effort
- Males usually compete for females
- **Sexual selection: selection for traits solely concerned with increasing mating success**

- **Intrasexual** → within our sex - competing for females
- **Intersexual** → Females choosing which male



→ Scooping out
seeds from
a previous
reproductive event

- **Males aren't choosy**, but may search for signals of fecundity



Adaptive Significance of Female Physical Attractiveness: Role of Waist-to-Hip Ratio

Devendra Singh

signals of fecundity

↳ Waist to hip
ratio
↳ Across cultures
preference for BMI varies,
but constant preference
for 0.7 waist to hip
ratio

Large breasts and narrow waists indicate high reproductive potential in women

Grażyna Jasieńska^{1*}, Anna Ziomkiewicz¹, Peter T. Ellison², Susan F. Lipson² and Inger Thune^{3,4}



Female reproductive strategy & mate preference

- High investment & focused on parental investment



- **Females “choosy”**

- Non-genetic benefits:
good resources and parental ability
- Genetic benefits

more
relevant
to humans





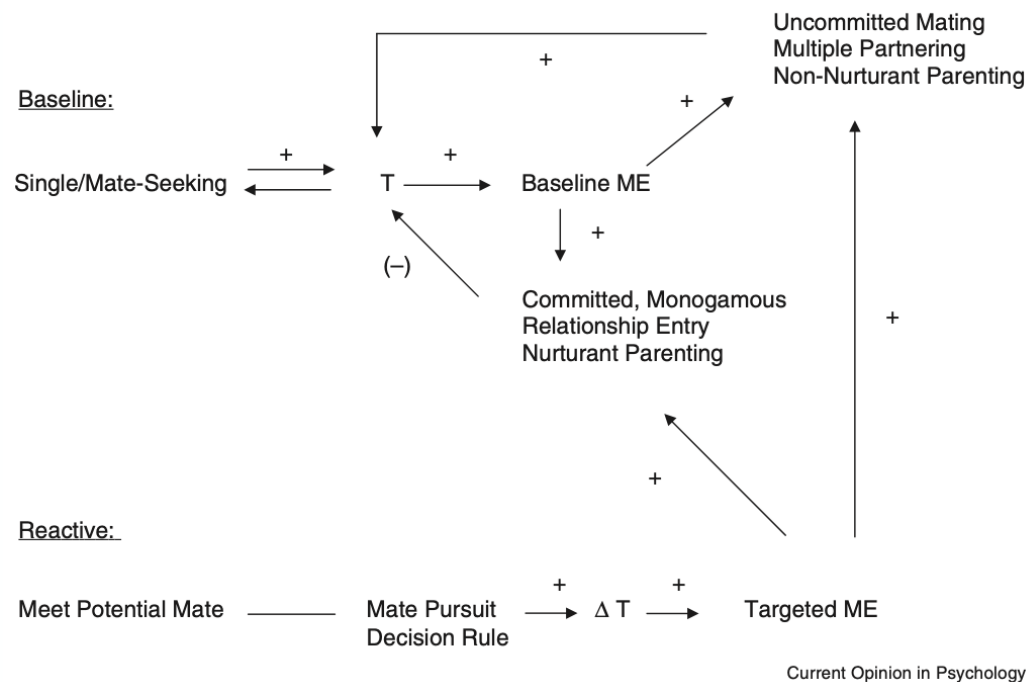
The inverse relationship between T and parental investment

High level is
of testosterone
↳ Higher mating
effort, lower
parental investment



So females
look for signals
of low testosterone

The Testosterone-Relationship Cycle



High T

Low T



- Non-genetic benefits:
good resources and parental ability

Intro

Reproductive strategies

Mate choice

Other



Honest signals

to of 'good genes'

- **Zahavi (1975): handicap hypothesis**
- Hamilton & Zuk (1982): reliable indicators of genetic resistance to disease
- *In humans: symmetry & genetic quality*

Facial symmetry is preferred



- Genetic benefits





Expectations for mate choice preferences in humans

	Males	Females
Gametes	Low	High
Parental investment	Comparatively high	Very high
Limiting factors	Females and resources	Resources

Small scale
Societies more
monogamous →
Stable pair bonds
↳ Humans evolved
within small societal
structure

Expectations:

- Trade-off between mating and parenting effort; balance must be sought in context of needy human babies
- Under stable reproductive bonds, males should find signals of fecundity important

Expectations:

- Needy babies: very little effort into mating, and lots into parenting
- Look for signals of
 - Willingness to invest in offspring
 - Resource access



Expectations for mate choice preferences in humans

We are about to conduct an experiment to test whether humans adhere to the patterns of mate preferences we expect based on differences in reproductive biology.

To recap, we expect males to pay attention to signals of **fertility** (e.g., waist-to-hip ratio), and we expect females to pay attention to **willingness to invest in offspring** (e.g., male access to resources, physical signals of low testosterone).

Our hypothesis is that males and females will differ in the extent to which they emphasize

1. Physical characters versus non-physical characters
2. Physical characters that convey what that sex should look for versus non-physical characters that relate to childrearing or investment in offspring

Expectations:

- Trade-off between mating and parenting effort; balance must be sought in context of needy human babies
- Under stable reproductive bonds, males should find signals of fecundity important

Expectations:

- Needy babies: very little effort into mating, and lots into parenting
- Look for signals of
 - Willingness to invest in offspring
 - Resource access



Expectations for mate choice preferences in humans

Our hypothesis is that males and females will differ in the extent to which they emphasize

1. Physical characters versus non-physical characters
2. Physical characters that convey what that sex should look for versus non-physical characters that relate to childrearing or investment in offspring

Male sample:

	Subject 1	Subject 2	Subject 3	Subject 4	Subject 5	Total
Physical characters						
Physical characters that may convey fertility ('pretty', 'attractive', etc)						
Any non-physical characters						
Non-physical characters that relate to childrearing or investment in offspring						



Expectations for mate choice preferences in humans

Our hypothesis is that males and females will differ in the extent to which they emphasize

1. Physical characters versus non-physical characters
2. Physical characters that convey what that sex should look for versus non-physical characters that relate to childrearing or investment in offspring

Female sample:

	Subject 1	Subject 2	Subject 3	Subject 4	Subject 5	Total
Physical characters						
Physical characters that may convey good genes or low T						
Any non-physical characters						
Non-physical characters that relate to childrearing or investment in offspring						

Mate choice preferences in humans

We are about to conduct an experiment to test whether humans adhere to the patterns of mate preferences we expect based on differences in reproductive biology.

To recap, we expect males to pay attention to signals of **fertility** (e.g., waist-to-hip ratio), and we expect females to pay attention to **willingness to invest in offspring** (e.g., male access to resources, physical signals of low testosterone).

Our hypothesis is that males and females will differ in the extent to which they emphasize

1. Physical characters versus non-physical characters
2. Physical characters that convey what that sex should look for versus non-physical characters that relate to childrearing of investment in offspring

Male sample:

	Subject 1	Subject 2	Subject 3	Subject 4	Subject 5	Total
Physical characters				Beams in		
Physical characters that may convey fertility ('pretty', 'attractive', etc)	Cute	Handy		Body		
Any non-physical characters	Being lazy w/		Simple		Secure come home	
Non-physical characters that relate to childrearing or investment in offspring		Knowing desires + commit + invest		Disciplined		

Female sample:

	Subject 1	Subject 2	Subject 3	Subject 4	Subject 5	Total
Physical characters					Looks are nice but thin	
Physical characters that may convey good genes or low T	Kind eyes					
Any non-physical characters	Respect	makes me laugh fun	smiley woman passion	Funny/fac	flow they try out people	
Non-physical characters that relate to childrearing or investment in offspring	Not self- absorbed	Good nature figure to kids likes kids wants to take care of children	alcohol- ability			



Expectations for mate choice preferences in humans

When poll is active respond at [PollEv.com/lauravanholstein822](https://poll-ev.com/lauravanholstein822)



Expectations:

- Trade-off between mating and parenting effort; balance must be sought in context of needy human babies
- Under stable reproductive bonds, males should find signals of fecundity important

Expectations:

- Needy babies: very little effort into mating, and lots into parenting
- Look for signals of
 - Willingness to invest in offspring
 - Resource access



Expectations for mate choice preferences in humans



Expectations:

- Trade-off between mating and parenting effort; balance must be sought in context of needy human babies
- Under stable reproductive bonds, males should find signals of fecundity important



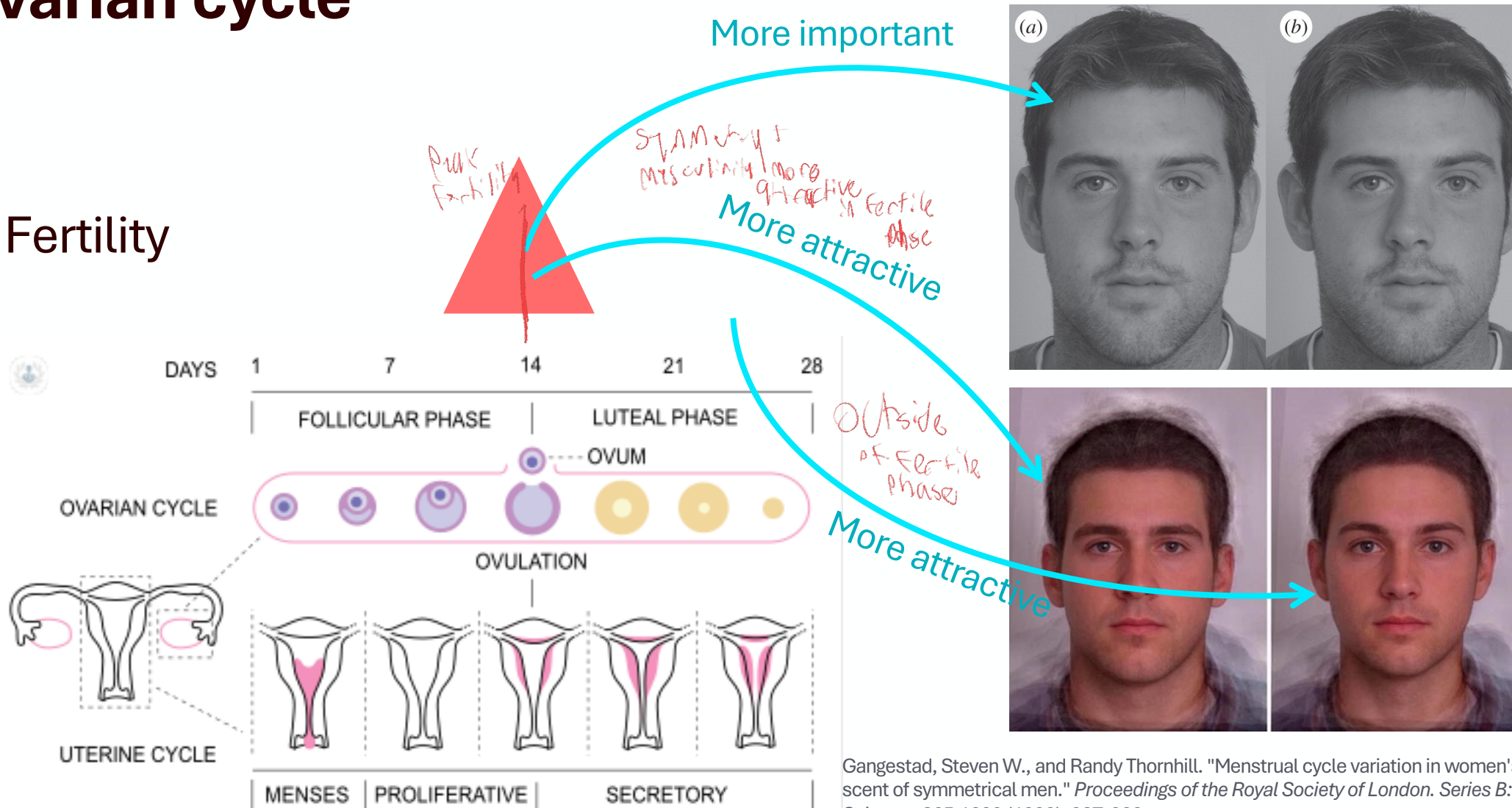
Expectations:

- Needy babies: very little effort into mating, and lots into parenting
- Look for signals of
 - Willingness to invest in offspring
 - Resource access



A major confound: reproductive interests across the ovarian cycle

Fertility



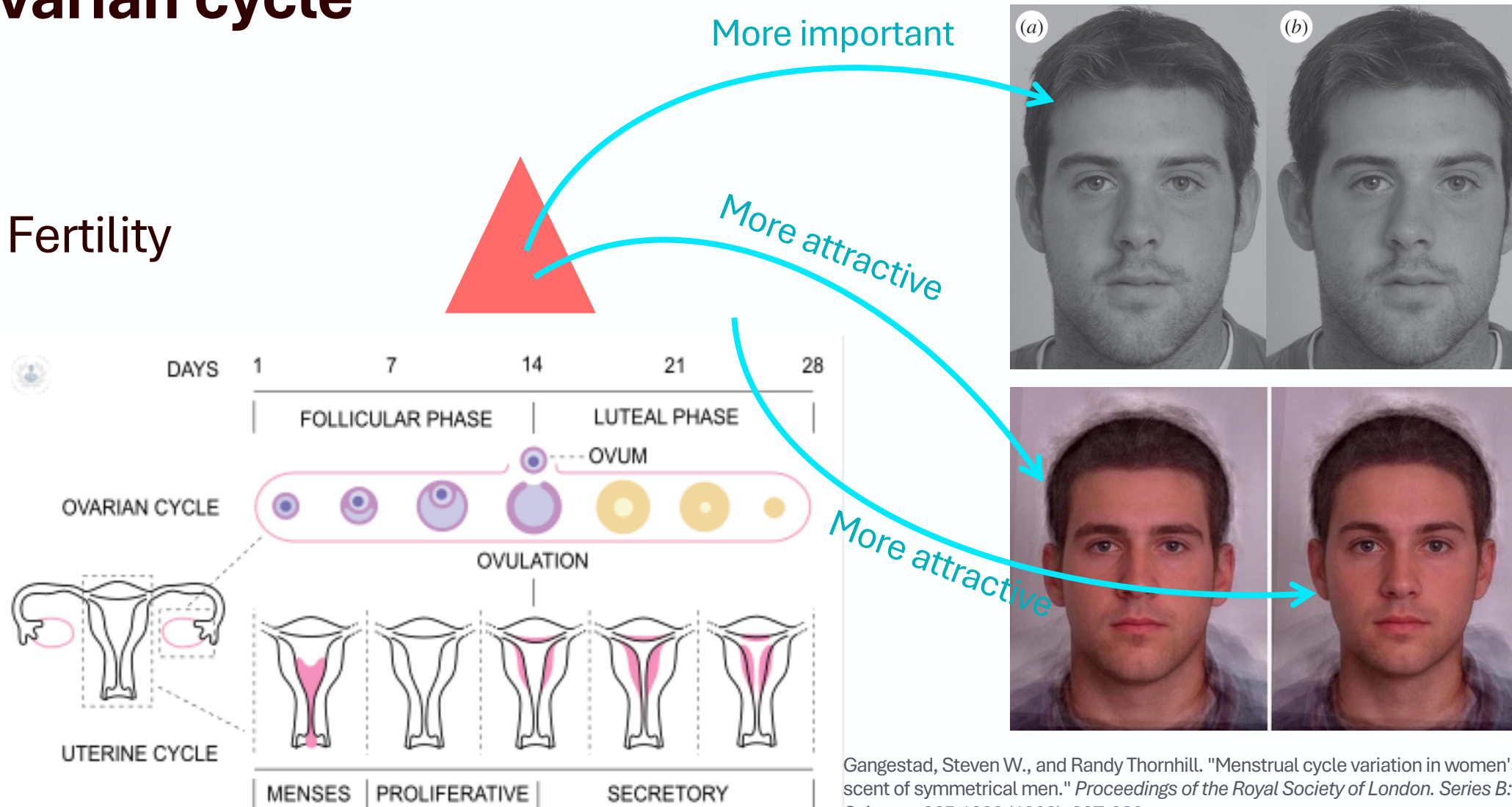
Gangestad, Steven W., and Randy Thornhill. "Menstrual cycle variation in women's preferences for the scent of symmetrical men." *Proceedings of the Royal Society of London. Series B: Biological Sciences* 265.1399 (1998): 927-933.

Jones, Benedict C., et al. "Effects of menstrual cycle phase on face preferences." *Archives of sexual behavior* 37.1 (2008): 78-84.



A major confound: reproductive interests across the ovarian cycle

Fertility



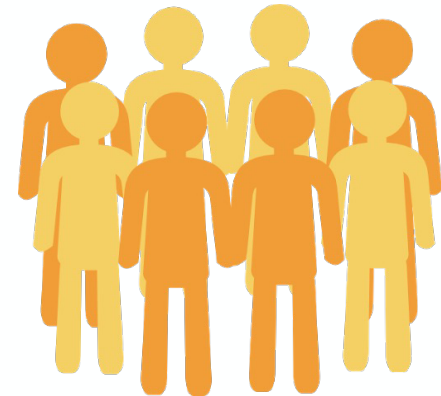
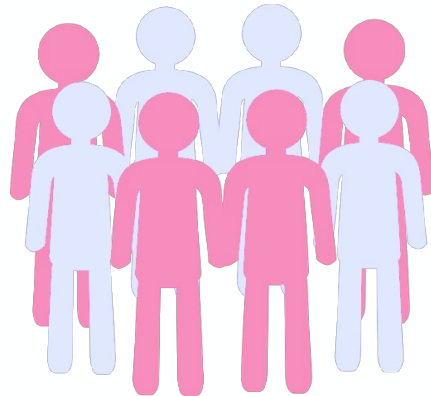
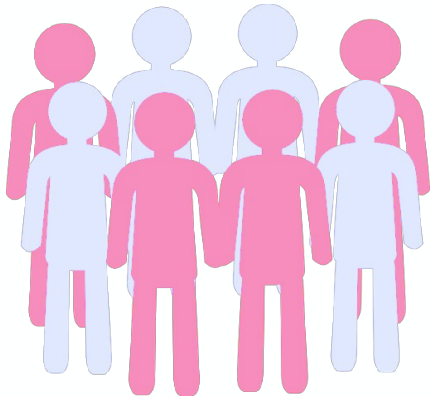
Gangestad, Steven W., and Randy Thornhill. "Menstrual cycle variation in women's preferences for the scent of symmetrical men." *Proceedings of the Royal Society of London. Series B: Biological Sciences* 265.1399 (1998): 927-933.

Jones, Benedict C., et al. "Effects of menstrual cycle phase on face preferences." *Archives of sexual behavior* 37.1 (2008): 78-84.



A major confound: culture

Preference
for BMI
Varies

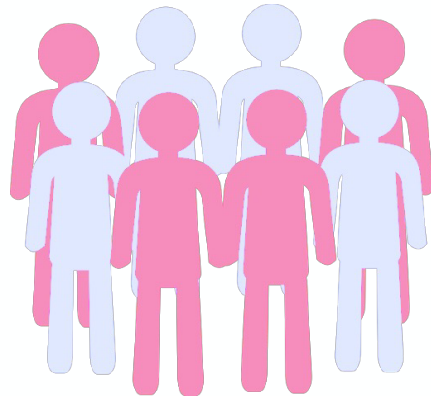
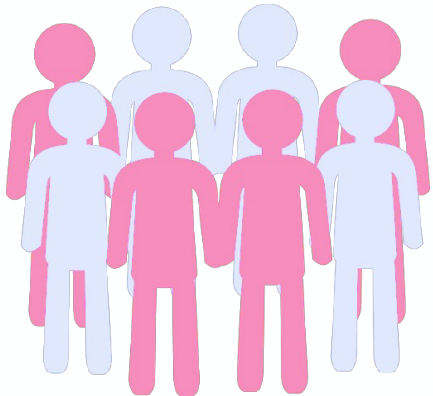




A major confound: culture

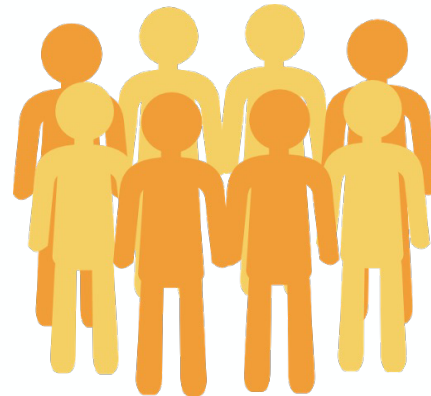
Buss (1989): Examined sex differences in mate choice in 37 populations

Across cultures

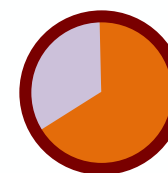
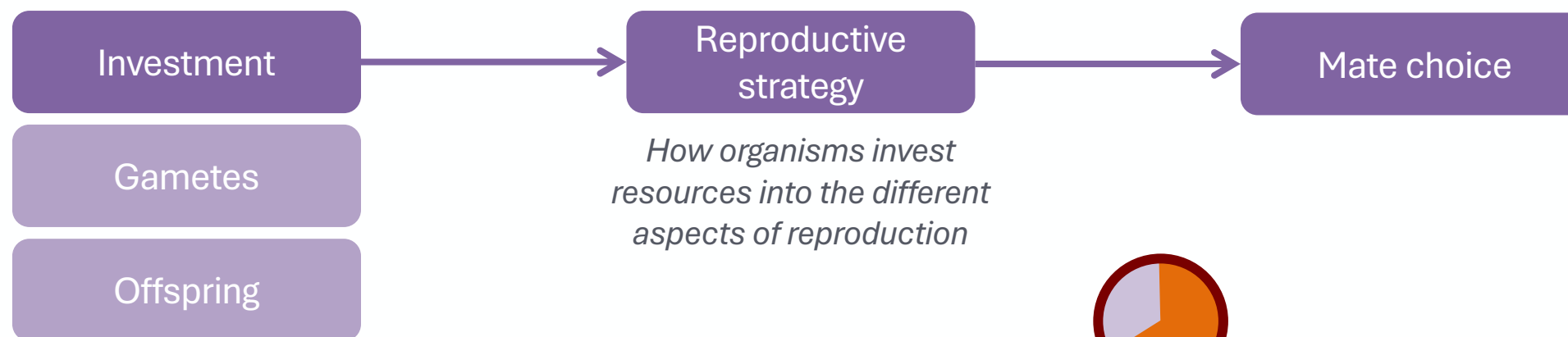


- In all 37 populations men rated physical attractiveness as more important; and in 36/37 women rated financial prospects as more important

What does cultural variation show



- Cultural variation found in relative importance of each variable tested



Parental investment

Mating effort



Take-home messages:

- Broadly, human averages conform to expectations, but...
 - The degree to which they do so is culturally dependent
 - Lots (and lots and lots) of variation
 - Lots of caveats



Reproductive
strategy

*How organisms invest
resources into the different
aspects of reproduction*

Frequency of
reproductive
attempts

Investment into
each offspring

Total number of
offspring

Duration of
pregnancy

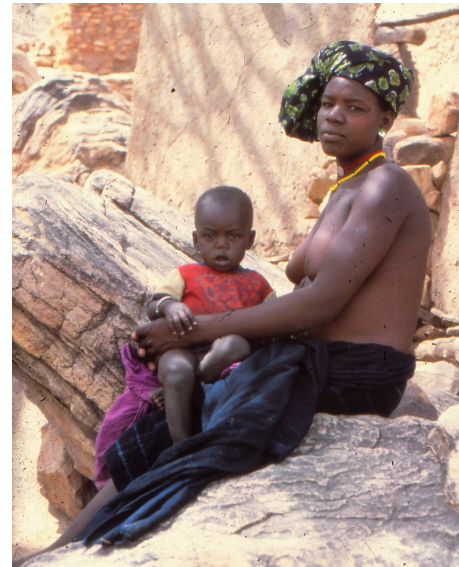
Duration of
breastfeeding

Different strategies

Among the Dogon (Mali, West Africa):

Trade-off per offspring:

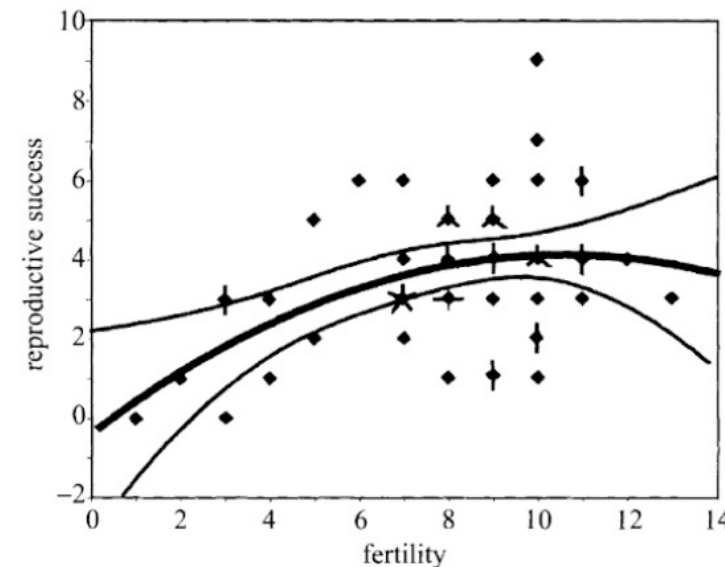
- + Reproductive success
- Chance of child mortality increases by 26% with each extra child in the nuclear unit



Total number of
offspring

Results:

- Max lifetime RS of 4.3 offspring obtained at fertility of 8.6 births.
- 75% of females achieved RS within CI of max.
- **83% of females had fertility level not significantly different from optimum**



Take-home messages:

Broadly, human *averages* conform to expectations based on our model relating reproductive traits to physiology, but...

- The degree to which they do so is **culturally dependent**
- Lots (and lots and lots) of variation within the sexes and between cultural groups
- There are many caveats. Studies often done on WEIRD populations, take humans at one moment in their lives (or menstrual cycles), and so on. Study design often doesn't allow for variability!